



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
20-06-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

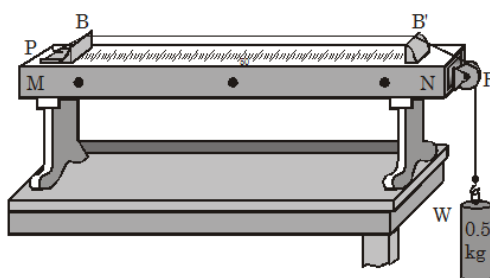
BEWARE OF NEGATIVE MARKING

PHYSICS

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The figure of a sonometer is shown here. The wooden box is calibrated such that the least count is 1 mm. The length of the sonometer wire BB' is 1.000 m and the hanging mass is 0.5kg. Take $g = 10 \text{ m/s}^2$. A wooden bridge is introduced in between B and B' such that the left part oscillates in 3 loops and right part in 5 loops. The frequency of tuning fork used is 500 Hz.



- (A) The bridge is at 37.5 cm from left end.
- (B) The speed of waves in the string is 125 m/s
- (C) The mass per unit length of the wire is 0.32 gm/m
- (D) We can shift the bridge by 25 cm and still we will obtain resonance.

Space for Rough Work

2. $M({}_{35}^{80}\text{Br}) = 79.918528\text{u}$, $M({}_{36}^{80}\text{Kr}) = 79.916376\text{ u}$, $M({}_{34}^{80}\text{Se}) = 79.916521\text{ u}$. Take mass of electron = $0.51\text{ MeV}/c^2$. $1\text{ amu} = 931\text{ MeV}$.
- (A) ${}^{80}\text{Br}$ can decay by β^- decay
- (B) ${}^{80}\text{Br}$ can decay by β^+ decay
- (C) ${}^{80}\text{Br}$ can undergo K shell capture.
- (D) ${}^{80}_{35}\text{Br}$ undergoes K shell capture so that ${}^{80}_{34}\text{Se}$ is formed.
3. We want to design a carbon resistor in shape of a cylinder. Its resistivity is known precisely. The diameter is measured by a screw gauge whose main scale has a least count of 1 mm and circular scale has 100 divisions. The length is measured by a vernier callipers whose main scale is 1 mm and 9 main scale divisions coincide with 10 vernier scale divisions. We wish that the fourth band in the colour code be painted gold. In which of the following screw gauge readings and vernier callipers readings will our goal be achieved?
- (A) Screw gauge: msr 2 divisions, 50th csd coinciding, Vernier scale: msr: 1.2 cm, 5th vernier division coinciding.
- (B) Screw gauge: msr 1 divisions, 0th csd coinciding, Vernier scale: msr: 0.3 cm, 3rd vernier division coinciding.
- (C) Screw gauge: msr 0 divisions, 40th csd coinciding, Vernier scale: msr: 5 cm, 0th vernier division coinciding.
- (D) Screw gauge: msr 1 divisions, 10th csd coinciding, Vernier scale: msr: 8 cm, 8th vernier division coinciding.

Space for Rough Work

4. A cuboid has sides of 10.4 cm, 3.8 cm and 1.02 cm. Choose the correct statement(s) about the area of the faces and its volume with due regard to significant digits.
- (A) The largest face area is 39.52 cm^2 .
(B) The smallest face area is 3.9 cm^2 .
(C) The volume of the cuboid is $4.0 \times 10^1 \text{ cm}^3$.
(D) The perimeter of the smallest face is 9.64 cm.
5. The plates of an isolated capacitor have equal and opposite charge of magnitude Q_0 . A positive insulator plate having same area as capacitor plates, and positive charge Q_0 is brought just outside the positive plate and kept parallel to it. Which of the following options is/are correct?
- (A) The electric field between the capacitor plates increases
(B) The outer surfaces of the capacitor plates have equal and opposite charges
(C) A net attractive force acts on the insulator plate due to the capacitor
(D) The net interaction force between the insulator plate and the negative plate of the capacitor is zero

Space for Rough Work

6. Two pin holes are made in a cardboard and monochromatic coherent plane wavefront is incident normally on the cardboard. The distance between the pin holes is very small compared to the observation screen. The cardboard is in x-y plane and the pin holes are located on the x axis symmetrically with respect to the origin. Choose the correct statement(s) from the following statements.
- (A) If the observation screen is parallel to x-y plane, the shape of fringes will be all straight lines.
- (B) If the observation screen is parallel to x-y plane, the shape of fringes will be hyperbola except the central line which will be straight and parallel to y axis.
- (C) If the observation screen is parallel to x-z plane, the shape of fringes will be all hyperbola except the central line which will be straight and parallel to y axis.
- (D) If the observation screen is parallel to y-z plane, the shape of fringes will be all semicircular.
7. An α particle with a kinetic energy of 2.1 eV makes a head on collision with a hydrogen atom moving towards it with a kinetic energy of 8.4 eV. As a result,
- (A) Collision may be elastic
- (B) Collision may be inelastic and a photon of energy 10.2 eV may be emitted
- (C) Collision will be elastic
- (D) Collision may be inelastic and a photon of energy 0.3 eV may be emitted

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 1 - 3

In a container fitted with a piston, we have one mole of a gas or a gaseous mixture whose total quantity is one mole. The composition of the gas is indicated in column-1. In column-2, the equation of the process carried out on the gas is mentioned. The column-3 mentions the pressure of the gas at the end of the process. The initial state of the gas in all the cases is pressure $P_0 = 10^5$ Pa and volume $V_0 = 25$ litre. Take $R = 25/3$ J/mol K.

Column-1		Column-2		Column-3	
(I)	Monoatomic gas	(i)	$P = \text{Constant}$	(P)	$P = 10^5$ Pa
(II)	Diatomic gas	(ii)	$V = \text{constant}$	(Q)	$P = 2 \times 10^5$ Pa
(III)	Mixture of monoatomic gas and diatomic gas in molar ratio 1:1	(iii)	$V = \text{constant}$	(R)	$P = 0.5 \times 10^5$ Pa
(IV)	Non linear polyatomic gas	(iv)	$PV^{-1} = \text{constant}$	(S)	$P = 4 \times 10^5$ Pa

- In which of the following combinations, heat is given to the gas but its temperature is decreasing?
 (A) (I) (i) (P) (B) (I) (iii) (R) (C) (II) (iv) (R) (D) (III) (iv) (R)
- In which of the following combinations, no heat is given to the gas or extracted from it?
 (A) (I) (iii) (Q) (B) (II) (iii) (S) (C) (III) (iii) (Q) (D) (IV) (iv) (Q)
- In which of the following combinations, the temperature of the gas increases?
 (A) (I) (i) (P) (B) (II) (ii) (R) (C) (III) (iii) (R) (D) (IV) (iv) (S)

Space for Rough Work

Paragraph for Questions 4 - 6

A square wire frame PQRS of side 1 m carrying a constant current of 4 A. The current is from $P \rightarrow Q \rightarrow R \rightarrow S$. In column-1, we have certain configurations producing a magnetic field. In column-2, we have mentioned the orientation of the loop and the plane of the loop. Column-3 indicates the axis about which the loop is free to rotate.

Column-1		Column-2		Column-3	
(I)	A long wire carrying a constant current of 1 A along positive z-axis	(i)	Loop is kept in xy plane with sides parallel to x and y axis. Current in wire PQ is in direction of positive x axis. Loop is in first quadrant.	(P)	Line PQ is the axis of rotation
(II)	An infinite sheet kept in xy plane carrying uniform current per unit length of 1 A/m flowing along x-axis	(ii)	Loop is kept in xz plane with sides parallel to x and z axis. Current in wire PQ is in direction of positive z axis. Loop is in first quadrant.	(Q)	Line QR is the axis of rotation
(III)	A long solenoid carrying current of 1 A and having 1000 turns/m. The axis of solenoid is z-axis. The loop is entirely inside the solenoid.	(iii)	Loop is kept in xy plane with sides parallel to x and y axis. Current in wire PQ is in direction of positive x axis. Loop is in second quadrant.	(R)	The loop is free to rotate about any axis.
(IV)	A small magnetic dipole of dipole moment 4 Am at the origin such that its dipole moment is along z-axis.	(iv)	Loop is kept in xz plane with sides parallel to x and z axis. Current in wire PQ is in direction of positive z axis. Loop is in second quadrant.	(S)	The diagonal PR is the axis of rotation

4. In which of the following cases, the loop does not rotate at all?
 (A) (I) (i) (S) (B) (II) (iv) (R) (C) (III) (ii) (P) (D) (IV) (ii) (Q)
5. In which of the following cases, the loop experiences a net force?
 (A) (I) (ii) (R) (B) (II) (i) (R) (C) (III) (i) (R) (D) (II) (iii) (R)
6. In which of the following cases, the loop starts rotating?
 (A) (II) (i) (P) (B) (II) (iii) (Q) (C) (III) (i) (P) (D) (III) (iii) (P)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
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- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

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Zero Marks : 0 In all other cases.

1. In a hydrogen like atom an electron is orbiting in an orbit having quantum number n . Its frequency of revolution is found to be 13.2×10^{15} Hz. Energy required to free the electron from the atom from the above orbit is 54.4 eV. In time 7 nano - second the electrons jumps back to orbit having quantum number $\frac{n}{2}$. τ be the average torque acted on the electron during the above process, then find $\frac{\tau}{3} \times 10^{27}$ in N – m.
(Given $\frac{h}{\pi} = 2.1 \times 10^{-34}$ J-s) Frequency of revolution of electron in the ground state of H - atom $f_0 = 6.6 \times 10^{15}$ Hz and ionization energy of H - bond $E_0 = 13.6$ eV
2. In a sample initially there are equal number of atoms of two radioactive isotopes A and B. Three days later nuclides of isotope A are half in number as compared to B, the half life of which is 1.5 days. What is the half-life (in days) of isotope A?
3. A proton is fired from very far away towards a fixed nucleus with charge $Q = 120 e$, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is : (take the proton mass. $m_p = (5/3) \times 10^{-27}$ kg; Round off to nearest integer.)

Space for Rough Work

4. A river is flowing with a speed of 5 m/s. A swimmer carries a tuning fork of frequency 110 Hz which he always manages to keep above water. The swimmer swims with a velocity of 6 m/s relative to water in a direction perpendicular to the river flow. An observer standing on the opposite bank also carried an identical tuning fork. Find the beat frequency (in Hz) heard by him corresponding to the sound emitted when the swimmer just begins to swim. The speed of sound in air is 336 m/s.
5. How many significant digits are in the answer of calculation $(2.31 - 0.31) \times 1632$

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 28)

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following statement is correct against to the indicated properties?

(A) $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_2 > \text{K}_2[\text{PtCl}_4]$; Crystal field splitting energy

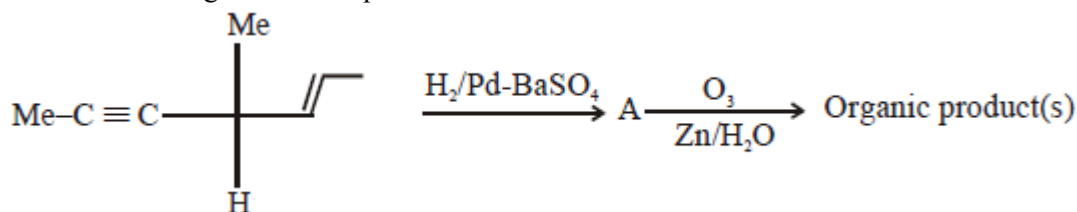
(B) $[\text{Co}(\text{NH}_3)_3\text{ClBr}(\text{NO}_3)] > [\text{Co}(\text{NH}_3)_3\text{Cl}_3]$; Number of geometrical isomers

(C) $\text{K}_4[\text{Fe}(\text{CN})_6] > \text{K}_3[\text{Fe}(\text{CN})_6]$; Valence shell's electrons of central metal ion

(D) $[\text{Fe}(\text{CO})_2(\text{NO})_2] > [\text{Fe}(\text{edta})]^\ominus$; Number of Fe-N linkages.

Space for Rough Work

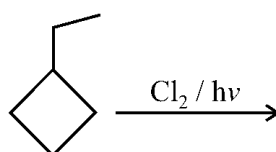
2. Consider following reaction sequence :



Which of the following statements are **INCORRECT** ?

- (A) Product A is optically inactive.
 (B) Three different types of organic products are formed in last step
 (C) Atleast one of the organic product formed in last step have aldehyde as a functional group
 (D) Compound A when treated with $\text{H}_2(\text{excess})/\text{Pt}$, produces a compound which is optically active.

- 3.

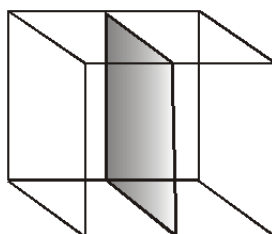


Correct statement about monochlorinated products of above reaction :

- (A) Total 10 monochlorinated products are formed
 (B) 6 resolvable products are formed
 (C) 4 enantiomeric pair are obtained
 (D) 7 fractions are obtained after fractional distillation of product mixture

Space for Rough Work

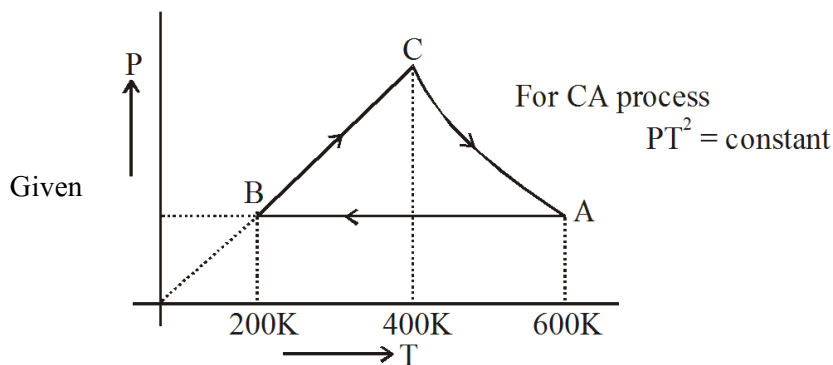
4. Select the correct statements
- (A) Catalyst can change the spontaneity of reaction
- (B) If n/p is higher than $\left(\frac{n}{p}\right)_{\text{stable}}$ then ${}_{-1}^0\beta$ particle may emit out
- (C) Binding energy per atom first increases then decreases with atomic mass
- (D) Rate of radioactive disintegration is independent of temperature
5. Solid AB form NaCl type structure in which A^+ is present at corner & face center of unit cell. If this unit cell is cut in two identical part according to given plane, these parts of unit cell called newly generated unit cell.



- (A) Empirical formula of AB remains unchanged
- (B) Unit cell formula (actual formula) of newly generated unit cell will be A_2B_2
- (C) Cation will be present on each face centres of newly generated unit cell
- (D) Newly generated unit cell will have same parameters that are of tetragonal lattice system

Space for Rough Work

6. Mark out the incorrect statement(s) [Atomic mass : S = 32]
- (A) At same temperature, average speed of O_2 is greater than that of SO_2
- (B) If absolute temperature of O_2 gas is half to that of SO_2 gas, then the speed distribution curves will be different
- (C) At given temperature, a gas with low molar mass has less K.E. than a gas with high molar mass
- (D) The real gas having higher value of Vander waal's constant "a" will be more compressible than one having lower value of "a", provided Vander waal's constant "b" is same for both the gases.
7. 1 mole of an ideal diatomic gas is subjected to following cyclic process.



which is/are correct statement(s) ?

- (A) Work done is zero in process BC.
- (B) Work done is +800 calories in process AB
- (C) Molar heat capacity of CA process is $11 \text{ cal mol}^{-1}\text{K}^{-1}$
- (D) Molar heat capacity of BC process is $6 \text{ cal mol}^{-1}\text{K}^{-1}$

Space for Rough Work

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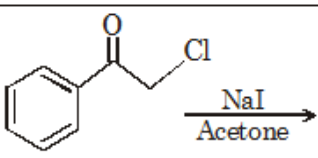
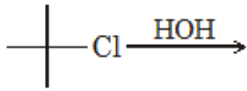
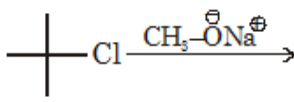
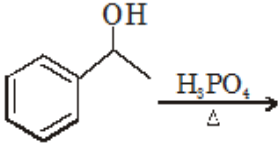
Paragraph for Questions 1 - 3

Column - I Number of Geometrical isomers	Column - II Number of optical active isomers	Column - III Pairs of Enantiomers
(I) 2	(i) 6	(P) 4
(II) 6	(ii) 3	(Q) 2
(III) 3	(iii) 4	(R) 3
(IV) 5	(iv) 2	(S) 1

- For $[\text{CrCl}_2(\text{en})(\text{NH}_3)_2]^{\oplus}$ complex the only correct combination is:-
 (A) (III), (iii), (P) (B) (I), (iv), (P)
 (C) (III), (iv), (S) (D) (III), (iv), (Q)
- For $[\text{PtCl}_2(\text{NH}_3)_2(\text{py})_2]^{+2}$ complex the only correct combination is :-
 (A) (II), (i), (S) (B) (II), (iii), (S)
 (C) (IV), (iv), (R) (D) (IV), (iv), (S)
- For $[\text{Co}(\text{gly})_3]^0$ complex the only correct combination is :-
 (A) (I), (iii), (R) (B) (I), (iii), (Q) (C) (I), (iv), (Q) (D) (II), (iii), (Q)

Space for Rough Work

Paragraph for Questions 4 - 6

	Column-I	Column-II	Column-III
(P)		(A) Major product is via E^1	(X) Product gives electrophilic substitution reaction
(Q)		(B) Major product is via E^2	(Y) D.B.E. of product is 5
(R)		(C) Major product is via S_N1	(Z) Product give electrophilic addition reaction with HOH, H^+
(S)		(D) Major product is via S_N2	(W) Product is aromatic

4. Correct match regarding (P) is :

- (A) C , Y , Z , W (B) D , Y , W (C) B , Z , W (D) C , Y , W

5. Correct match regarding (R) is :

- (A) C,Y (B) B,Y (C) B , Z (D) A , Z

6. Correct match regarding (S) is :

- (A) B , Y , Z (B) C , W (C) D, Y , Z (D) A , Y , Z , W

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

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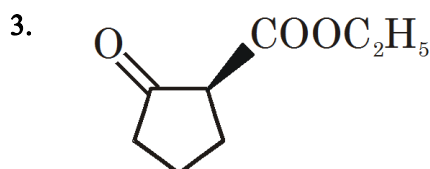
Zero Marks : 0 In all other cases.

1. Consider following complexes :
- $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_2(\text{NH}_3)]$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Zn}(\text{gly})_2]$, $[\text{Co}(\text{H}_2\text{O})_3(\text{NH}_3)_3]$, $[\text{Pt}(\text{en})\text{Br}_3\text{Cl}]$, $[\text{Co}(\text{gly})_3]$

How many of these show either geometrical or optical isomerism but not both ?

2. How many of them are sulphide ore ?

Iron Pyrite, Chalcopyrite, Galena, Chalcocite, Magnetite, Cinnabar, Sphalerite, Alumina, Siderite, Calamine, Malachite.



When the above molecule is reduced with excess LiAlH_4 followed by dilute acid hydrolysis, how many organic products can be isolated by fractional distillation of product mixture.

Space for Rough Work

4. Find (O – O) bond enthalpy in H_2O_2 (kJ/mol)

Given : $\Delta H_f(\text{H}_2\text{O}_2, l) = -200 \text{ kJ/mol}$

$\Delta H_f(\text{H}_2\text{O}, l) = -285 \text{ kJ/mol}$

$\Delta H_{\text{vap.}}(\text{H}_2\text{O}_2, l) = 60 \text{ kJ/mol}$

$\Delta H_{\text{vap.}}(\text{H}_2\text{O}, l) = 40 \text{ kJ/mol}$

$\Delta H_{\text{atomisation}}(\text{O}_2, g) = 300 \text{ kJ/mol}$

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

5. The activation energy of a second order reaction : $\text{A} + \text{B} \rightarrow \text{P}$ is $9.2 \text{ kcal mole}^{-1}$. At 500K, product is formed at the rate of $0.08 \text{ mol L}^{-1}\text{min}^{-1}$. Find the rate of formation of product at 1000 K in $\text{mol L}^{-1}\text{min}^{-1}$, if other factors are assumed to be same [Use $\ln x = 2.3 \log x$]

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
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- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a strictly increasing function such that $x \cdot f(x) + 1 > 0$ and $f(x) \left[f\left(f(x) + \frac{1}{x}\right) \right] = 1$ for all $x > 0$, then which of the following is/are correct-

(A) $f(1) = 2 \sin\left(\frac{\pi}{10}\right)$

(B) $f(1) = 2 \sin\left(\frac{11\pi}{10}\right)$

(C) $\operatorname{sgn} f(x) = 1$ for $x > 0$

(D) $\operatorname{sgn} f(x) = -1$ for $x > 0$

2. A differentiable function satisfy the equation $f(x) = \int_0^x (f(t) \cos t - \cos(t-x)) dt$ then which of the following is/are true

(A) $f'(0) = -1$

(B) $f''\left(\frac{\pi}{2}\right) = e$

(C) $f'(0) = 1$

(D) $f''\left(\frac{\pi}{2}\right) = -e$

Space for Rough Work

3. Area bounded by $\sqrt{x+y+1} < \sqrt{y-x+2}$ & $y < 0$ is less than

(A) $\frac{9}{16}$

(B) $\frac{9}{5}$

(C) $\frac{5}{4}$

(D) $\frac{16}{5}$

4. If $I_m = \int_0^{m\pi} x |\sin x| e^{\cos 4x} dx$ and $J_n = \int_0^{n\pi} |\cos x| e^{\cos 4x} dx$, where $m, n \in \mathbb{I}$ then

(Note : $[k]$ denote GIF)

(A) $\left[\frac{I_1}{J_1} \right] = 0$

(B) $\left[\frac{I_2}{J_2} \right] = 3$

(C) $\left[\frac{I_2}{J_1} \right] = 0$

(D) $\left[\frac{I_3}{J_3} \right] = 4$

Space for Rough Work

5. The value of definite integral $\int_{-2014}^{2014} \frac{dx}{1 + \sin^{2015}(x) + \sqrt{1 + \sin^{4030}(x)}}$ equals
- (A) 0 (B) 2014 (C) $(2014)^2$ (D) 4028
6. If ' α ' is minimum value of $f(x) = \frac{x^4 - 2x^2 + 3}{(x^2 + 1)^2}$ and ' β ' is/are value(s) of x at which $f(x)$ attains its minimum, then
- (A) α is rational number (B) β is/are irrational number(s)
- (C) $\alpha = \frac{1}{3}$ (D) $|\beta| = \sqrt{2}$
7. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the equation $f(x + y) = f(x - y) + f(x).f(y) \forall x, y \in \mathbb{R}$. Identify the correct statement(s)
- (A) $f(2014) + f(2015) = f(2016)$ (B) $\sin^{-1}(f(2)) + \cos^{-1}(f(3)) = \frac{\pi}{2}$
- (C) $f(2014) - f(2015) = f(2016)$ (D) $\sin^{-1}(f(3)) + \cos^{-1}(f(2)) = \frac{\pi}{2}$

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 1 - 3

Let $f(x) = \frac{(x-a)(x-b)}{(x-c)(x-d)}$ where $a < b < c < d$. Column-I: contains information about real roots of the equation; Column-II: contains information about limit of $f(x)$; Column-III: contains information about increasing interval and decreasing interval of $f(x)$ and $f'(x)$ and information about local minimum value and local maximum value of $f(x)$

Column-I		Column-II		Column-III	
(I)	$f(x) = 1$ has exactly one real root	(i)	$\lim_{x \rightarrow c^-} f(x) = -\infty$	(P)	$f(x)$ is increasing in the interval (b, c)
(II)	$f(x) = 2$ has exactly two real root	(ii)	$\lim_{x \rightarrow d^+} f(x) = \infty$	(Q)	$f(x)$ is decreasing in the interval (d, ∞)
(III)	$f(x) = -10$ has one real root	(iii)	$\lim_{x \rightarrow d^-} f(x) = -\infty$	(R)	$f(x)$ is increasing in the interval (b, c)
(IV)	$f(x) = 10$ has two real root	(iv)	$\lim_{x \rightarrow \infty} f(x) = 1$	(S)	local minimum value of $f(x)$ is smaller than local maximum value of $f(x)$

- Which of the following options is the only **CORRECT** combination ?
 (A) (I) (ii) (S) (B) (II) (iii) (Q) (C) (I) (i) (R) (D) (III) (iv) (S)
- Which of the following options is the only **INCORRECT** combination ?
 (A) (II) (ii) (P) (B) (III) (i) (R) (C) (IV) (iv) (P) (D) (I) (iii) (Q)
- Which of the following options is the only **CORRECT** combination ?
 (A) (III) (iv) (P) (B) (I) (ii) (R) (C) (IV) (i) (S) (D) (II) (iv) (S)

Space for Rough Work

Paragraph for Questions 4 - 6

Let $f(x)$ be a double differentiable function with continuous second derivative defined on the interval $[0, 1]$ satisfying the equation $\int_0^x \sqrt{1 - (f''(t))^2} dt = \int_0^x f'(t) dt \forall x \in [0, 1]$ and $f(0)=f'(0)=0$. Column-I:

contains the information about interval of value of $f(x)$; Column-II: contains information about value of $f(x)$; Column-III: contains information about interval of value of definite integral of $f(x)$; Match the following Column(s)

Column-I		Column-II		Column-III	
(I)	$f\left(\frac{1}{2}\right) < \frac{1}{8}$	(i)	$f\left(\frac{1}{3}\right) = \frac{1}{18}$	(P)	$\int_0^1 f(x) dx > 1$
(II)	$f\left(\frac{1}{2}\right) > \frac{1}{8}$	(ii)	$f\left(\frac{1}{4}\right) = \frac{1}{8}$	(Q)	$\int_0^1 f(x) dx > \frac{1}{6}$
(III)	$f\left(\frac{1}{3}\right) < \frac{1}{18}$	(iii)	$f\left(\frac{1}{6}\right) = 2\sin^2\left(\frac{1}{12}\right)$	(R)	$\int_0^1 f(x) dx < \frac{1}{6}$
(IV)	$f\left(\frac{1}{4}\right) < \frac{1}{4}$	(iv)	$f\left(\frac{1}{8}\right) = 2\sin^2\left(\frac{1}{16}\right)$	(S)	$\int_0^1 f(x) dx < \frac{1}{4}$

4. Which of the following options is the only **CORRECT** combination ?
 (A) (I) (ii) (Q) (B) (II) (i) (S) (C) (IV) (ii) (Q) (D) (III) (iii) (R)
5. Which of the following options is the only **CORRECT** combination ?
 (A) (I) (iv) (Q) (B) (II) (iii) (S) (C) (II) (iv) (P) (D) (IV) (iii) (S)
6. Which of the following options is the only **CORRECT** combination ?
 (A) (III) (ii) (Q) (B) (IV) (ii) (P) (C) (I) (iii) (R) (D) (III) (ii) (S)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories:
Full Marks : +3 If ONLY the correct answer is chosen.
Zero Marks : 0 In all other cases.

1. If $f : \mathbb{R} - \{0,1\} \rightarrow \mathbb{R}$ be a function satisfying $f(x) + f\left(1 - \frac{1}{x}\right) = \tan^{-1}x$ and $N = f\left(\frac{1}{2}\right)$, then the value of $\left(\frac{8N}{\pi}\right)$ is
2. If $\int e^x \left(\frac{x+2}{x+4}\right)^2 dx = e^x \left(\frac{x-\alpha}{x+\beta}\right) + C$ (where C is integration constant), then $|\beta - \alpha|$ is equal to
3. If $\sqrt{x+y} + \sqrt{y-x} = 5$, then $25 \frac{d^2y}{dx^2}$ is equal to

Space for Rough Work

4. The value of the expression $\lim_{y \rightarrow \infty} ((8y+3)(y+1)(y+2)(y+3)(2y+3)(4y+5)(2y+7))^{1/7} - 2y$ is of the form $\frac{A}{B}$ (Where A, B are coprime number). Then the value $\left[\frac{A}{B} \right]$ (where $[.]$ denotes the greatest integer function), is
5. Number of points, where $f(x) = |x| + \left| \frac{x}{3} - 1 \right| + \left| x - \left| \frac{x}{3} - 1 \right| \right|$ is non derivable, is

Space for Rough Work

Space for Rough Work